

3-Mavzu: Pythonda tarmoq dasturlashga kirish.

5-mashg'ulot. PyQt5 paketidan foydalanib zamonaviy chat dasturini tuzish.

O'quv savollari:

1. PyQt5 orqali TCP client uchun GUI yaratish.
2. Socket ulanishini boshqarish (connect/send/receive) va xabar almashish.
3. Signal-slot mexanizmi va xatoliklarni GUI'da ko'rsatish.

Mashg'ulot maqsadi:

PyQt5 yordamida TCP client uchun grafik interfeys (GUI) yaratish, socket orqali xabar almashish, signal-slot mexanizmini o'rganish va GUI'da xatoliklarni ko'rsatish.

1. PyQt5 orqali TCP Client uchun GUI yaratish

Asosiy GUI tuzilmasi

Quyidagi dasturda PyQt5 yordamida **chat client interfeysi** yaratiladi:

```
import sys
import socket
from PyQt5.QtWidgets import QApplication, QWidget, QVBoxLayout, QHBoxLayout,
QTextEdit, QLineEdit, QPushButton, QLabel
from PyQt5.QtCore import Qt, QThread, pyqtSignal
```

Chat oynasini yaratish

```
class ChatClient(QWidget):
    def __init__(self):
        super().__init__()
        self.setWindowTitle("TCP Chat Client")
        self.setGeometry(300, 200, 400, 400)

        # Asosiy layout
        layout = QVBoxLayout()

        # Xabarlar oynasi
        self.chat_box = QTextEdit()
        self.chat_box.setReadOnly(True)
        layout.addWidget(QLabel("Chat oynasi:"))
        layout.addWidget(self.chat_box)

        # Xabar yuborish qismi
        hbox = QHBoxLayout()
        self.input_box = QLineEdit()
        self.send_button = QPushButton("Yuborish")
        hbox.addWidget(self.input_box)
        hbox.addWidget(self.send_button)
        layout.addLayout(hbox)
```

```
self.setLayout(layout)

# Tugma bosilganda signal
self.send_button.clicked.connect(self.send_message)
```

Izoh:

- QVBoxLayout — elementlarni ustma-ust joylashtiradi.
- QLineEdit — matn kiritish uchun.
- QTextEdit — chat xabarlarini ko'rsatadi.
- QPushButton — "Yuborish" tugmasi.

2. Socket Ulanishini Boshqarish (connect / send / receive)

Socket bilan ishlash

Quyida TCP client uchun socket funksiyasi berilgan:

```
class ClientSocket(QThread):
    message_received = pyqtSignal(str)    # Signal: serverdan xabar kelganda
    error_signal = pyqtSignal(str)        # Signal: xatolik bo'lsa

    def __init__(self, ip, port):
        super().__init__()
        self.server_ip = ip
        self.server_port = port
        self.client_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
        self.running = True

    def run(self):
        try:
            self.client_socket.connect((self.server_ip, self.server_port))
        except socket.error as e:
            self.error_signal.emit(f"Ulanishda xatolik: {e}")
            return

        while self.running:
            try:
                data = self.client_socket.recv(1024)
                if not data:
                    break
                self.message_received.emit(data.decode('utf-8'))
            except:
                self.error_signal.emit("Server bilan aloqa uzildi.")
                break

    def send_message(self, msg):
        try:
            self.client_socket.sendall(msg.encode('utf-8'))
        except:
            self.error_signal.emit("Xabar yuborishda xatolik yuz berdi!")
```

```
def stop(self):
    self.running = False
    self.client_socket.close()
```

Izohlar:

- QThread yordamida tarmoq operatsiyalari GUI'ni muzlatmasdan fon rejimida bajariladi.
- PyQtSignal yordamida **signal-slot mexanizmi** ishlatiladi (quyida tushuntiriladi).

GUI bilan socketni bog'lash

Endi ChatClient sinfini ClientSocket bilan bog'laymiz:

```
class ChatClient(QWidget):
    def __init__(self):
        super().__init__()
        self.init_ui()

        # Server ma'lumotlari
        self.client = ClientSocket("127.0.0.1", 5000)
        self.client.message_received.connect(self.display_message)
        self.client.error_signal.connect(self.display_error)
        self.client.start()

    def init_ui(self):
        self.setWindowTitle("TCP Chat Client")
        self.setGeometry(300, 200, 400, 400)

        layout = QVBoxLayout()
        self.chat_box = QTextEdit()
        self.chat_box.setReadOnly(True)
        layout.addWidget(self.chat_box)

        hbox = QHBoxLayout()
        self.input_box = QLineEdit()
        self.send_button = QPushButton("Yuborish")
        hbox.addWidget(self.input_box)
        hbox.addWidget(self.send_button)
        layout.addLayout(hbox)

        self.send_button.clicked.connect(self.send_message)
        self.setLayout(layout)

    def send_message(self):
        msg = self.input_box.text().strip()
        if msg:
            self.client.send_message(msg)
            self.chat_box.append(f"Siz: {msg}")
            self.input_box.clear()
```

```
def display_message(self, msg):
    self.chat_box.append(f"Server: {msg}")

def display_error(self, error):
    self.chat_box.append(f"<span style='color:red;'>Xatolik: {error}</span>")
```

Dastur ishga tushirish

```
if __name__ == "__main__":
    app = QApplication(sys.argv)
    window = ChatClient()
    window.show()
    sys.exit(app.exec_())
```

3. Signal–Slot Mexanizmi va Xatoliklarni GUI’da Ko‘rsatish

Signal–Slot tushunchasi

PyQt5 da **signal–slot** mexanizmi komponentlar o‘rtasida aloqa o‘rnatish uchun ishlatiladi.

- **Signal** — bu voqea sodir bo‘lganini bildiradi (masalan, tugma bosildi yoki xabar keldi).
- **Slot** — bu voqea sodir bo‘lganda bajariladigan funksiya.

Masalan:

```
self.send_button.clicked.connect(self.send_message)
```

bu joyda `clicked` signali `send_message()` slotiga ulanadi.

GUI’da xatolikni ko‘rsatish

Tarmoqdagi xatolik foydalanuvchiga GUI oynasida **qizil rangda** ko‘rsatiladi:

```
def display_error(self, error):
    self.chat_box.append(f"<span style='color:red;'> Xatolik: {error}</span>")
```

Misol natijasi:

1. Klient oynasi ishga tushadi.
2. Foydalanuvchi xabar yozadi va “Yuborish” tugmasini bosadi.
3. Serverdan javob keladi.
4. Aloqa uzilganda GUI’da xabar chiqadi:
5. Xatolik: Server bilan aloqa uzildi.

Xulosa

O‘rganilgan tushunchalar	Tavsif
PyQt5 GUI	Grafik interfeys yaratish uchun ishlatiladi
Socket	Tarmoq orqali ma’lumot yuborish va olish
QThread	Fon jarayonlarda tarmoq ishlarini bajarish
Signal–Slot	GUI komponentlar o‘rtasida aloqa
Xatoliklarni ko‘rsatish	GUI’da foydalanuvchiga qulay tarzda ogohlantirish berish

